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ChE 210 Quiz #3 Tuesday 17 March 2026

To Receive Full Credit, You Must Show All Your Work

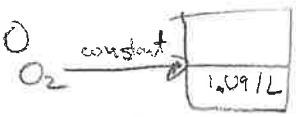
Ethanol is converted to acetic acid by the reaction $C_2H_5OH + O_2 \rightarrow CH_3COOH + H_2O$. The process takes place in a batch reactor initially filled with an aqueous solution of ethanol with concentration 1.09 mol/L. A stream of O_2 is passed through the reactor to maintain a constant concentration of dissolved O_2 in the reactor throughout the process. The reaction rate is second order in the ethanol concentration: $r_A = kC_A^2V$ where $C_A = n_A/V$ and V is assumed to be constant at 2500 L. The subscript A indicates the ethyl alcohol. Ignore any volume change caused by the production of water.

(a) If the reaction rate r_A has units of mol/day, and C_A has units of mol/L, what are units for the constant k?

(b) Write a mole balance for ethanol. Combine this with the reaction rate.

(c) Use the result of part (b) to calculate the time to convert 85% of the ethanol to acetic acid. The numerical value of k is 0.01156 with the appropriate units from part (a).

$C_2H_5OH + O_2 \longrightarrow CH_3COOH + H_2O$



a) r_A = mol/day $\quad C_a$ = mol/L $\quad V = L$

$r_A = k C_a^2 V$

$$\frac{mol}{day} = k \frac{mol^2}{\cancel{L^2}} \cdot \cancel{L}$$

$$\frac{mol}{day} = k \frac{mol^2}{L} \qquad k = \frac{L}{\cancel{mol} \cdot day} \cdot \frac{mol^2}{\cancel{L}} = \frac{mol}{L}$$

$$\boxed{k = \frac{L}{mol \cdot day}}$$

b) $\int n_{Af} - n_{Ai} = \int \underset{\text{mi}}{\frac{1.09 mol}{L}} + \underset{r_A}{k C_A^2 V} \qquad \nu_A = -1$

$$\boxed{n_{A,sys} = \int_{t_0}^{t_f} 1.09 mol/L - k(1.09 mol/L)^2 \cdot 2500 L}$$

c) $\underset{n_{Ai}}{100 \text{ mols of ethanol}} \xrightarrow{1:1 \text{ ratio}} \underset{n_{Af}}{85 \text{ ethanol}}$ to 85 acetic acid

$$-15 = \Big[1.09t - 34.33t\Big]_0^{t_f}$$

$$\cancel{-}15 = \cancel{-}33.24t^f$$

$$\boxed{t = 0.451 \text{ day}}$$

NAME ______________________________

Some useful integrals:

$$\int (a+bx)dx = \frac{1}{2b}(a+bx)^2$$

$$\int \frac{dx}{(a+bx)} = \frac{1}{b}\ln|a+bx|$$

$$\int \frac{dx}{x(a+bx)} = -\frac{1}{a}\ln\left|\frac{a+bx}{x}\right|$$

$$\int \frac{x\,dx}{(a+bx)} = \frac{1}{b^2}(a+bx-a\ln|a+bx|)$$

$$\int \frac{dx}{(a+bx)^2} = \frac{-1}{b(a+bx)}$$

$$\int \frac{(A+Bx)}{(a+bx)}dx = \frac{(bA-aB)\ln|a+bx| + bBx}{b^2}$$

$$\int \frac{dx}{(a+bx)(A+Bx)} = \frac{1}{(bA-aB)}\ln\left|\frac{a+bx}{A+Bx}\right|$$